DATA ANALYTICS / LINKEDIN PROJECT CASE STUDY

ANTENATAL CARE UTILIZATION ANALYSIS

Cross-sectional statistical research using SPSS and Excel



RESEARCH FOCUS

Determinants of adequate antenatal care utilization among pregnant women in Bangladesh

5,064	40.8%	0.696
Women analyzed	Adequate ANC	ROC AUC

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Research Overview

Study design, objectives and analytical framework

5,064	40.8%	59.2%
Observations	Adequate ANC	Inadequate ANC

Research question

Which factors are associated with adequate antenatal care utilization among pregnant women in Bangladesh?

Study design

Cross-sectional secondary analysis. Binary outcome: adequate ANC = 1 and inadequate ANC = 0.

Tools and methods

SPSS output interpretation, Excel reporting, descriptive statistics, cross-tabulation, chi-square testing, binary logistic regression and ROC analysis.

Interpretive scope

The analysis identifies associations in the observed data. Cross-sectional comparisons cannot establish causal relationships.

Analytical Workflow

From dataset review to statistical interpretation

- Review the study objectives, variable coding and sample size.
- Summarize the respondent characteristics and adequate ANC distribution.
- Use cross-tabulations and chi-square tests for bivariate associations.
- Estimate adjusted associations using binary logistic regression.
- Assess model classification, pseudo-R-squared, Hosmer-Lemeshow and ROC AUC.
- Translate statistical output into findings and research recommendations.

Descriptive and Bivariate Results

Education and wealth group differences

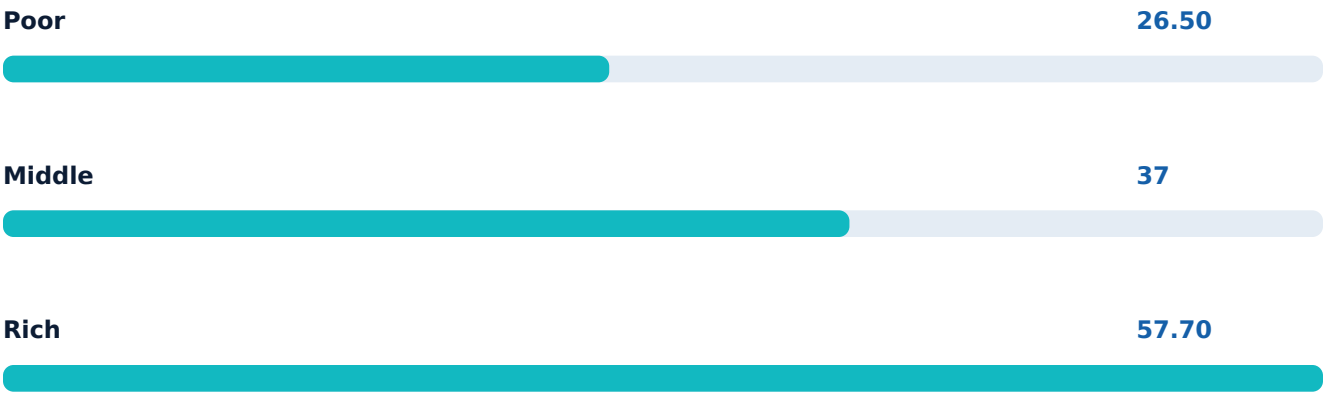
Adequate ANC by education

Observed adequate ANC: low education 35.3%; high education 64.5%. Chi-square = 275.035; p < 0.001.



Adequate ANC by household wealth

Poor: 26.5%; middle: 37.0%; rich: 57.7%. Chi-square = 414.657; p < 0.001.



Interpretation: higher education and wealth groups show higher observed adequate ANC rates; these results do not prove causation.

Logistic Regression Results

Selected adjusted odds ratios from the source report

Age group of woman

Adjusted OR 1.242 | 95% CI 1.097–1.405 | p 0.001

Type of residence

Adjusted OR 0.631 | 95% CI 0.552–0.721 | p <0.001

Women’s education

Adjusted OR 1.632 | 95% CI 1.370–1.943 | p <0.001

Wealth index

Adjusted OR 1.484 | 95% CI 1.374–1.602 | p <0.001

67.0%

Classification accuracy

0.696

ROC AUC

0.168

Nagelkerke R^2

Model fit: omnibus chi-square = 672.465 ($p < 0.001$); Hosmer-Lemeshow $p = 0.060$. Odds ratios must be interpreted in light of the original variable coding.

Findings and Research Implications

Evidence-based interpretation and next steps

- Adequate ANC utilization was reported for 40.8% of the analyzed women.
- Education and household wealth were associated with different adequate ANC rates.
- Selected adjusted predictors include age group, residence, education and wealth index.
- The ROC AUC of 0.696 indicates moderate discrimination in the reported model.
- Future analysis should check coding, model specification and potential confounding before drawing policy conclusions.
- This case study demonstrates statistical reporting and interpretation rather than a causal evaluation.